Learning Video-Audio Alignment Through Multimodality Representation

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Abstract—Multimodal alignment between audio and video streams is a fundamental requirement for media integrity, yet existing systems lack robust methods for verifying whether these modalities are temporally consistent. In this work, we investigate whether transformer-based multimodal representations can be used to automatically detect video–audio misalignment. Using the VS13 vehicle dataset, we construct an augmented corpus of aligned and deliberately misaligned video–audio pairs by applying controlled temporal shifts to the audio track. We then propose a dual-encoder architecture that combines Video Vision Transformer (ViViT) and Audio Spectrogram Transformer (AST) embeddings, fused through a two-layer fully connected classifier trained to distinguish aligned from misaligned pairs. Our experiments demonstrate that the model successfully learns cross-modal correspondences and achieves strong classification performance despite the small dataset size, highlighting the effectiveness of transformer-based representations for temporal alignment verification. An ablation study confirms the importance of embedding dimensionality and classifier depth in capturing multimodal relationships. This work provides a step toward generalizable, automated tools for audiovisual integrity verification and lays the groundwork for future extensions such as continuous offset estimation and multi-view or multi-sensor alignment.

Index Terms—audio, classification, machine learning, alignment, multimodality.

I. Introduction

A. Motivation

The proliferation of digital multimedia has led to audio and video data being shared, distributed, and streamed across the internet at unprecedented rates, with video alone constituting a majority of global internet traffic. Driven by platforms like YouTube, which alone accounts for 16% of all internet traffic in 2024 [1] (see Fig. 1), underscoring the dominant role of audiovisual content online. However, this proliferation of audio-visual content has introduced critical challenges in content authenticity and integrity. The rise of deepfake technology and sophisticated audio-video manipulation techniques poses significant threats to information security, public trust, and individual safety—from election interference and financial fraud to personal defamation. Despite advances in detecting manipulated media, a fundamental gap remains: existing verification systems lack robust methods for assessing the semantic and temporal alignment between audio and video streams in real-time. This project is motivated by this landscape and aims to address the question: Can we robustly and automatically verify the temporal alignment between a video feed and its corresponding audio track? Our proposed model aims to identify inconsistencies between audio and visual information that may indicate tampering, thereby contributing to the broader ecosystem of media authentication and helping safeguard against the societal harms of synthetic media manipulation.

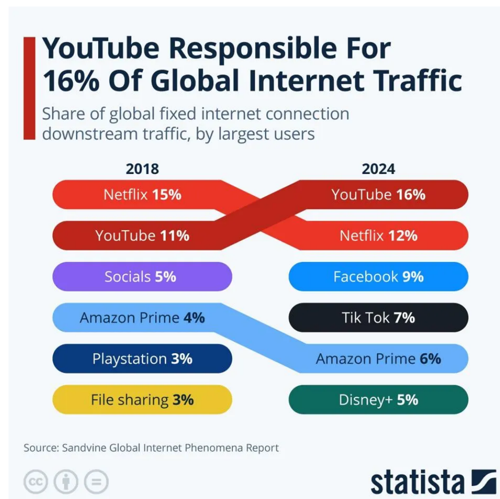

Fig. 1: Share of global fixed internet connection downstream traffic, by largest users[1]

B. Related Work

We first reviewed the literature to understand what is the current state of the art, what techniques are used, and which types of models have been applied on such task.

Dosovitskiy et al. [2] introduces the first Vision Transformer (ViT) which attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.

Arnab et al. [3] shows how to effectively regularize the model during training and leverage pretrained image models to be able to train on comparatively small datasets, and achieves state-of-the-art results on multiple video classification benchmarks.

Gong et al. [4] introduces the first convolution-free, purely attention-based model for audio classification, which achieves new state-of-the-art results on various audio classification benchmarks.

Radford et al. [5] demonstrates that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet.

Girdhar et al. [6] shows that all combinations of paired data are not necessary to train a joint embedding, and only image-paired data is sufficient to bind the modalities together, and that ImageBind serves as a new way to evaluate vision models for visual and non-visual tasks.

II. Proposed Approach

Our approach is divided into two primary procedures. Utilizing the VS13 dataset from Djukanović et al. [7], we first create an augmented dataset and assign new labels depending on if our newly generated examples are aligned. We then utilize our augmented dataset to train and fine-tune our model to identify whether a pair of video-audio tracks are aligned. We detail these approaches in the following sections.

A. Dataset Preprocessing

We use the dataset in Djukanović et al. [7]. This dataset is composed of 400 ten-second video and audio recordings collected from 13 different vehicles. All recordings were captured using a single GoPro Hero5 Session positioned at a fixed viewpoint. Vehicle speeds range from 30 to 105 km/h. In each video sequence, the vehicle maintains a constant velocity. The camera was placed 0.5 m from the road at a height of 1.2 m to ensure consistent viewpoint geometry across the dataset. Each video is capture on a near identical, straight course with some variance in the background. See an example frame from the dataset video data in Fig. 2.

Originally the VS13 dataset was designed for road vehicle speed estimation. We adapted this dataset for our alignment tasks. The data has been processed in the following way. We cropped both video and audio files to capture the middle 4 seconds of each 10-second clip. This timing captures the key moment when the vehicle passes the camera and disappears from the frame around the 5-second mark in the original videos. This also roughly aligns with the peak audio.

Fig. 2: Example frame from the dataset video

To create misaligned video-audio pairs, we shifted the audio forward or backward by 2 seconds. To keep consistency from video to video we decided not the shift the video file like the audio files. We figured this consistency would help with the learning process and make it easier to understand if video is our baseline modality. See the schema in Fig. 3.

For each vehicle, we created a 50/50 split. Half with correct video audio alignment and half with forward or backward 2 second audio offset.

B. Model Architecture and Training

Our model architecture consists of two encoder models which feed into a fully connected network for classification, as shown in Fig. 4. The two encoders are tasked with producing a hidden representation of the video track and audio tracks. The encoders hidden representations for the video and audio track are concatenated together and then used as input into the fully connected layer. In our study, we employ two transformer-based architectures, Video Vision Transformer (ViViT) from Arnab et al. [3] and Audio Spectrogram Transformer (AST) from Gong et al. [4], for video and audio representation, respectively. Primarily, we use these transformer models for their inherent deep learning capabilities and out-of-the-box pre-trained parameters to produce high quality representations for the video and audio tracks. In the following sections, we go into detail about these specific transformer models, their suitability for our task, their use in the fully connected classification head, and our training process.

1) Video Vision Transformer (ViViT): Video Vision Transformer (ViViT) [3] is a transformer-based vision model designed to process and classify video data. Drawing inspiration from its predecessor Vision Transformer (ViT) from Dosovitskiy et al. [2], ViViT extracts spatio-temporal tokens from the input video utilizing ViT's attention-based image patching strategy. After extracting these spatio-temporal tokens, ViVit then applies its own self-attention mechanism on the sequence of tokens constituting a video, capturing long-range dependencies between tokens in time

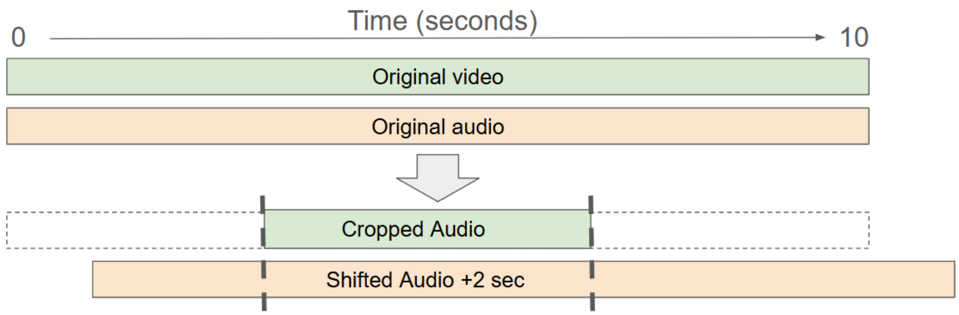

Fig. 3: Data processing schema for a single video-audio pair with a offset of +2 seconds

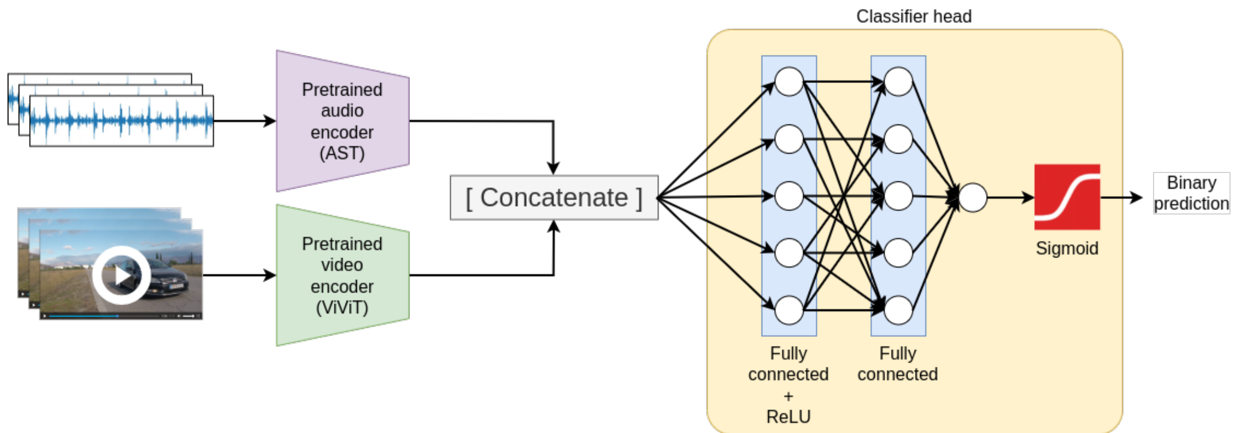

Fig. 4: Model architecture

of the video and creating high quality contextualized hidden representations of the video. ViViT has been shown to outperform other state-of-the-art deep 3D convolutional networks on video classification benchmarks such as Kinetics 400 and Epic Kitchens.

An important consideration in our selection of ViViT for our video encoder is the size of our dataset. While we process the data to produce our ground truth data examples from the original dataset, we only end up with approximately 400 data points, which is considered a small dataset for Deep Learning models. In approaches utilizing transformer-based architectures, a large dataset is necessary to be able to successfully fine-tune the transformer on the target task, especially in the situation of a domain switch from the pre-training data to the target task data. Moreover, video-based datasets such as ours are often smaller than conventional text-based or tabular datasets which poses an issue for applying a transformer-based models on video data, in general. However, ViViT incorporates various regularization techniques and initializes its video models with pretrained image models trained over a wide range of domains. These two strategies effectively allow ViViT to be fine-tuned on our dataset, despite the size of our dataset.

As mentioned previously, we use ViViT to extract a hidden representation of the video alongside the hidden representation of the audio for input into our fully connected layer. We accomplish this by obtaining a tensor representation of the 120 frames (4 second video @ 30 FPS), consisting of shape (120, 1920, 1080, 3) where the first dimension is the number of frames (sequence length), the second dimension is the width of the frame, the third dimension is the height of the frame, and the fourth dimension is the number of channels. ViViT processes this tensor for use in its attention mechanism as part of its pipeline. Our hidden representation of the video is the last-layer hidden state of the classification token normally used for video classification in the ViViT pipeline.

2) Audio Spectrogram Transformer (AST): Audio Spectrogram Transformer (AST) [4] is transformer-based approach to obtaining hidden representations for audio data. AST approaches audio embedding from a vision perspective similar to ViT [3]: it first takes the raw audio data and produces a spectrogram image of the audio and then uses the spectrogram image to create embeddings by applying a vision transformer. AST is a purely attention based model and can capture long-range dependencies and context throughout the audio track even at its lowest levels. AST has been shown to beat state-of-the-art CNN-attention hybrid models on audio benchmarks such as AudioSet, ESC-50, and Speech Commands.

Similar to the problem with training transformer models on video data, audio datasets are often smaller than what is normally expected for training transformer models for foundation models. AST takes a similar approach as ViViT in that they take a transfer learning approach by first transforming an audio track into a 2D audio spectrogram image and then apply a linear projection of the patched spectrogram image to produce a hidden representation for the audio track. ViT [2], as used in ViViT, and AST are similar in architecture but have two key differences: AST extends ViT by modifying the positional encoding of ViT to adapt to the spatio-temporal dependencies of the audio track through time; and AST utilizes the weights of the three RGB channels in ViT for its own patch embedding layer. This allows AST to be use a pre-trained ViT model for variable input shapes without having to adjust the architecture for various configurations of ViT.

We find AST to be an appropriate choice for our audio embedding as we aim to leverage its pre-trained knowledge about audio spectrogram images and the dependencies between audio through time to produce high quality embeddings for our audio as input into the fully connected layer alongside the video embedding. Unfortunately, the pre-training of AST focuses on speech and natural language recognition which means that our task is a domain shift for the model. However, our goal is not to extract the absolute best hidden representation for our audio track but instead to get a workable audio representation which the models can use to learn implicit features and relationships between the audio embedding and the video embedding. Moreover, we want to take advantage of AST's self-attention mechanism and its innate ability to capture long range dependencies as those features are likely to be important indicators of video-audio alignment.

3) Fully connected classification head: Our fully connected classification head consists of two linear layers and a ReLU activation layer between the linear layers (see Fig. 4). The FC head accepts the audio and video embeddings concatenated together. The first linear layer will take the FC input and output a vector twice the size of the concatenated audio and video embeddings and the second linear layer will take this intermediate vector and output a single classification value. Our notion in formulating of the classification head in this manner is that we want the head to learn the inter-dependencies between the video and audio embeddings. More succinctly, we want the classification head to learn features about pairwise vector values and embed information about video-audio alignment into its intermediate representation before outputting a final prediction.

4) Model Training: We train our model to identify whether or not pairs of audio and video are aligned, framing the task as binary classification. We split our dataset into training, validation, and test sets consisting of 70%, 15%, and 15% of the dataset, respectively. We train the model on the training set and evaluate the model's performance at each epoch on the validation set. After completing all epochs, we perform one final evaluation of the model on the test set. All training and evaluation is performed on an NVIDIA RTX 6000 Ada Generation. We utilize binary cross entropy loss as our learning objective and the Adam optimizer for optimization through gradient descent. We train the model for 20 epochs with a batch size of 50. We report the evolution of the loss function during training in Fig. 5. As expected, we can see an overall decreasing trend, but the loss does not converge. Training on video data was extremely time-consuming (i.e., one hour per epoch), which is why we decided to train the model only on 20 epochs. We would need to train on more epochs to see the loss reaching convergence. In Fig. 6, we report the evolution of the accuracy and F1 score computed on both the training and validation set during training. We can see an increasing trend in accuracy and F1 score during training. We note that, thanks to pretraining, performance is already high at the first epoch of training. Here we report only one run of training, but we should note that these numbers can have a lot of variance between runs, probably due to the randomness of the weights initialization.

Batching our dataset for mini-batch gradient descent requires special care due to the constraints of the hardware used to train the model. Naively, one could convert all video/audio files in our dataset into vectors and batch the vectors. This would be resource-intensive as each converted and processed video and audio vector can be resource-heavy and require a large amount of memory. With around 400 video files and audio files, it is infeasible for us to load all the vectors for batching. We instead batch the paths to the video and audio files, forming them into processed vectors in an ad-hoc manner. This comes at the cost of computation time as we must perform vectorization and processing of these files for every batch before input into the model. However, we gain additionally flexibility as we can adjust the batch size and experiment with the limits of our hardware to determine the range of batch size which is appropriate for our system.

An important consideration to our training process and model implementation is hyperparameter selection and hyperparameter tuning. The performance of the transformer-based encoders of our models and the classification head depend on a good set of hyperparameters thus we make a substantial effort to ensure that our selected hyperparameters are as optimal as possible. We perform random hyperparameter search on five main hyperparameters: the size of our audio and video hidden representations which both have the same size; the number of hidden layers in the transformer networks; the number of attention heads in the transformer networks; the size of the intermediate representations use in the transformer network; and the learning rate. Our random hyperparameter randomly selects a set of values for each hyperparameter and trains a newly initialized model for one epoch, recording the final

Hyperparameter	Selected Value
Number of hidden layers	6
Number of attention heads	3
Intermediate representation size	300
Hidden representation size	300
Learning rate	0.0001

TABLE I: Our final selection of hyperparameters after 50 steps of random hyperparamter search.

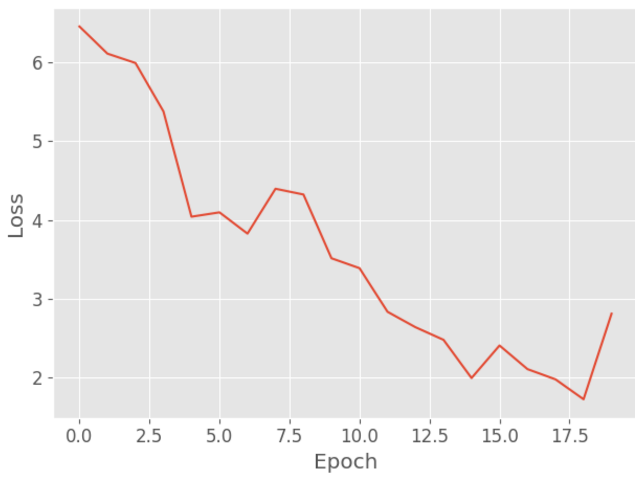

Fig. 5: Evolution of the loss function during training

validation loss. After n step, the hyperparameters which produced a model with the lowest final validation loss is the best selection of hyperparameters. Our experiments ran the search for 50 steps on a subset of 100 examples, for brevity. Our final selection of hyperparameters is summarized in Table I.

III. Experimental Results and Discussion

In this section, we present the experiments conducted to test if our model can successfully model a joint distribution of audio and video data. We discuss the metrics used to quantify the performance of our model on the task of identifying mismatched video-audio pairs, present our results on the task, and provide insight on our results. Additionally, we perform an ablation study to help support our initial findings and insights. Our model was trained on a single NVIDIA RTX 6000 Ada.

A. Evaluations and Metrics

We use four metrics to evaluate the performance of our model on the task of identifying mismatched video-audio pairs: accuracy, precision, recall, and F1 score. Our dataset is perfectly balanced, so we avoid any issues that may come from using accuracy as a metric on unbalanced datasets. However, to ensure a succinct evaluation of our model, we also compute precision, recall, and F1 score on the test set

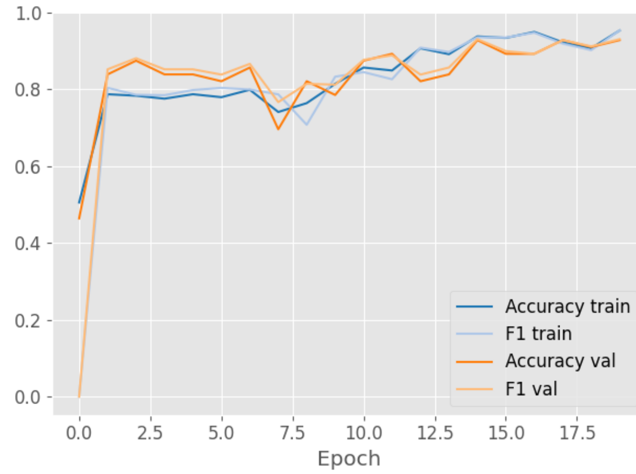

Fig. 6: Evolution of the accuracy and F1 score during training on the training set (blue) and on the validation set (orange)

predictions and ground truths to determine model aptness in detecting misaligned video-audio pairs. As precision and recall are components of F1, we utilize these metrics to explain how the model is performing, why it is performing as such, and the implications of the computed metrics on the model and the target task. Additionally, to get a deeper understanding of the performance of our model, we also look into the Area Under the Receiver Operating Characteristic (ROC) curve, and we report the confusion matrix on the test set.

B. Insight and Analysis

We present the results of our training and evaluation of the model on the task of video-audio misalignment detection in Table II. From our initial experiments, we see that our model is able to successfully detect whether a given audio and video pair is aligned with high performance. With an accuracy of 0.786 and an F1 score of 0.727, our model is able to assess the relationship between a video-audio pair by first forming them into machine-understandable features by way of the video and audio transformers, respectively, and then utilizing the innate relationships between these extracted features to make a correct prediction in the classification head.

To get a better understanding of the model predictions, we plot the confusion matrix in Fig. 7. On a total of 56 held out data examples in our test set, our model makes only 4 false positive and 3 false negative predictions.

In Fig. 8, we also look into the area under the ROC curve, which plots the true positive rate (TPR, or sensitivity) against the false positive rate (FPR) of our binary classifier. Roughly, the perfect classifier curve should be as close as possible to the top left corner of the plot. In our case, we see that the ROC curve follows the top left corner pretty closely, with an area under the curve of

Model configuration	Accuracy	Precision	Recall	F1
Base model	0.786	0.8	0.667	0.727
Hidden size=120	0.571	0.602	0.532	0.565
Hidden size=600	0.375	0.375	1	0.545
3-layer classifier head	0.536	0.536	1	0.698
1-layer classifier head	0.732	0.655	0.792	0.717

TABLE II: The results of experiments on training our model to detect video-audio misalignment, including the ablation study. From the varying hidden sizes, we see that fine tuning of the hidden size is important as too small of a hidden representation causes the model to lose information and too large of a hidden representation introduces too much noise into the representation. From the varying classification head layers, we see that more layers introduces additional noise causing classification performance to decline but that a single fully-connected layer is not enough to learn the relationship between video-audio pairs.

0.96. Since training didn't reach convergence (see Fig. 5) the performance could probably be further improved by training on more epochs.

The success of the base model in our video-audio alignment task can be attributed to the two main components of our model. First, both the video and audio transformers are able to successfully form hidden representations for the entire sequence of video and audio tracks which are highly representative of components that are most indicative of an aligned video and an aligned audio. The important notion with the optimization of our transformer-based encoders is that we are not necessarily producing the highest quality representations of our video and audio tracks through these encoders. In other words, the hidden representations of our audio and video tracks are unlikely to be useful in a transfer-learning scenario. This is because we adjust the parameters of these encoders using the gradient of the loss function catered to the specific task of video-audio alignment. Thus, our encoders learn what components of the video specifically indicate video audio alignment in our dataset, rather than a generalizable set of features to be used in a wider range of tasks. Nevertheless, the success of our transformer-based encoders can be attributed to their specific architectural designs. Notably, both ViViT [3] and AST [4] utilize pre-trained vision transformer models to facilitate their usage in small-scale datasets, such as ours. Furthermore, the use of an attention mechanism to ingest contextual tokens in both the video and audio sequences and their incorporation into the final hidden representation is instrumental to our model's success as these embeddings give our classification head contextual information about where a point in audio/video is located in time relative to other time points. This allows for our classification head to better recognize anomalies in our data (i.e., misaligned video-audio pairs) as this contextual information is often indicative of where a frame/audio sample is positioned in time.

Second, our classification head is able to learn the relationship between the audio-video pairs. We select a two-layer classification head (see Fig. 4) where our input

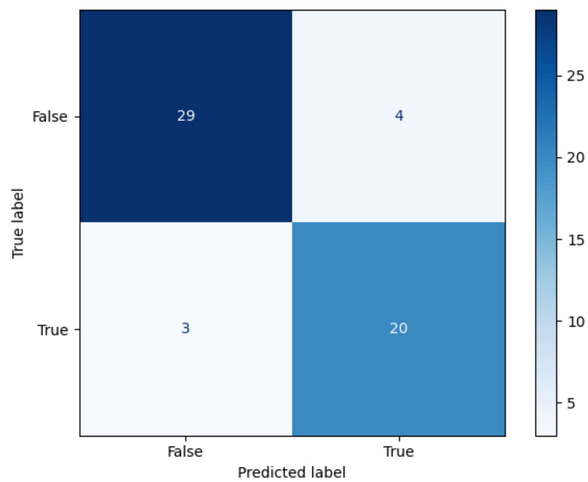

Fig. 7: Confusion matrix predictions on the test set

layer transforms the concatenated vectors into a vector double the size of the input to capture all interactions across all the hidden states of the input video and audio. Intuitively, this allows the classification head to process information about pairs of video-audio hidden states which it can use to determine whether or not the video and audio pairs are aligned. Furthermore, the backpropagation of the gradient directly to these parameters allows for further understanding of video-audio alignment features, which the classification head can better detect on subsequent forward passes through the model. We empirical show the positive outcomes of selecting this specific architecture for our classification head in our ablation study below.

C. Ablation Study

In this section, we detail our ablation study which was conducted to gain further insight on our initial results. With this, we aim to investigate two main assertions of our base models: (1) that a carefully selected hidden size is important for model performance; (2) that our specific

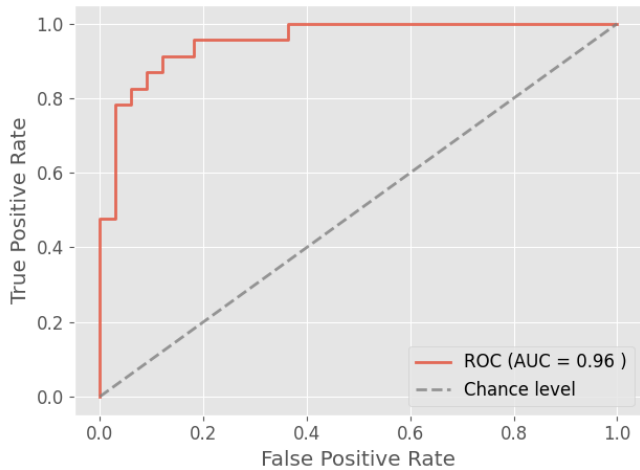

Fig. 8: Area under the ROC curve

choice of the depth of the classifier head is optimal for our task.

1) Choice of hidden size: In our first ablation study, we investigate whether the size of the audio and video representation produced by the transformers has any significant impact on the performance of the model. Maintaining the other hyperparameters and performing the same train-evaluation approach, we re-train the model to produce hidden representations of size 120 and 600 and pass these concatenated representations to the classification head. Our selection of these specific hidden sizes is deliberate, as 120 is near half and 600 is double the size of the original hidden representations. We present our results in Table II. From these results, we see that both halving the hidden size or doubling the hidden size causes a decline in model performance. For halving the hidden size, this can be explained by the model not receiving enough information about the video and audio sequences to successfully learn about how the pairs of their hidden representations indicate alignment. Interestingly, doubling the hidden size sees a sharper decline in model performance than halving the hidden size. We explain this by considering that adding additional dimensions (i.e., a bigger size) to the embedding space introduces additional noise by making the space more sparse. Thus, the classification head is unable to infer the relationship between the video and audio due to the sparsity of space, leading to the decline in classification performance.

2) Classifier head depth: In our second ablation study, we investigate whether the depth of the fully-connected classification head impacts the performance of the model. As mentioned, we make specific choice of architecture for the classification head to capture relations between all pairs of hidden representation values with the goal of incorporating information that is indicative of misaligned pairs during backpropagation. To show that this architecture is effective, we modify the classification head to have 1 layer and 3 layers, and we retrain the model with these new classification heads. Our results of these experiments are shown in Table II. Similar to our results with adjusting the hidden size, we see that adjusting the number of layers in the fully-connected classification head causes model performance to decline. With the 3-layer classification head, the additional layer adds a additional gradient that must be calculated and backpropagation during optimization. This likely causes the more important earlier layer of the model to lose important signals that indicate whether the video-audio pairs are aligned. The 1-layer classification head also performs worse than our base model but comparably less than the 3-layer classification head. Nevertheless, the inclusion of the second layer in our base model allows the head to learn relationships between pairs of hidden representation values, an aspect that proves to be important as visible by the decline in performance in its exclusion in the 1-layer classification head.

IV. Conclusions and Future Work

In this work, we present a multimodal framework for classifying video-audio pair alignment using transformer-based encoders for both modalities. Using the VS13 dataset, we were able to create a balanced dataset that allowed us to benchmark the performance of our model. In our approach we used ViViT for video representation and AST for audio representation, followed by a fully connected classification head to learn cross-modality connections.

Through our experiments, our model demonstrated that it was able to make correct audio-video pair alignment classification. In our ablation study we also showed that appropriate hidden dimensionality and two-layer classification head are important for maximizing performance. Overall, our model have demonstrated to be able to capture patterns across multi-modalities.

A natural direction for future work would be to move beyond binary alignment classification and predict continuous temporal misalignment between audio and video. Estimating the magnitude of the offset would give the possibility, not only to detect misalignment, but also have the capability to correct it. This extension would directly address the original safety motivation from Section I-A, particularly for applications such as autonomous vehicles, where even small cross-modal delays can pose risks.

Another extension would be to develop a more generalizable model. Our current dataset is limited to a single viewpoint, which does not reflect the variability encountered in real-world deployments. Expanding this preliminary study to include multiple viewpoints and a wider range of vehicle trajectories and orientations would improve robustness and scalability.

Finally, it would be interesting to incorporate additional modalities, such as IMU or depth data, to further enhance the model's ability to reason about multimodal alignment and improve performance in complex environments.

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